MATERIAL SAFETY DATA SHEET

Version：V1.0

PA-3223

1. Identification	
1.1 Product name:	Aqueous acrylic dispersion PA-3223
Chemical category:	Aqueous acrylic dispersion
1.2 Product use suggestions and restrictions:	Used in coatings.
1.3 Supplier information:	
Name of supplier:	Guangzhou Guanzhi New Material Technology Co., Ltd.
Supplier address:	Room 1106, Area A, Guangzhou International Business Incubator, No. 3 Juquan Road, Science City, Luogang District, Guangzhou
Tel:	86-20-82567990
Fax:	86-20-32214789

2.Hazards Identification

GHS hazard classification: Not classified as a hazardous chemical according to GHS.

Hydroxy acrylic polymer GHS classification: Not applicable;

Label elements:

Please note the following substance:

N,N-Dimethylethanolamine, neutralizing agent, bonded as salt, mass concentration < 2.0%, specific concentration limit >= 5%

GHS classification:

Flammable liquid (Category 3), H226;

Acute toxicity, Inhalation (Category 4), H332;

Acute toxicity, Dermal (Category 4), H312;

Skin corrosion (Category 1B), H314;

Specific target organ toxicity - single exposure (Category 3), H335;

Specific target organ toxicity - single exposure (Category 3), H335;

GHS pictograms: None

Signal word: Warning

Hazard statements: No known significant effects or critical hazards.

Precautionary statements:

Prevention:

2.Hazards Identification

Wear protective gloves, eye protection and protective clothing during operation.

Wash contact areas thoroughly after handling.

Do not eat, drink or smoke when using this product.

Use only outdoors or in a well-ventilated area.

Response:

Inhalation: If inhaled, get medical advice/attention if you feel unwell.

Skin and eye contact: Wash thoroughly with soap and water. If irritation persists, seek medical attention.

Ingestion: Rinse mouth immediately. Do NOT induce vomiting. Immediately seek medical advice/attention.

Spillage: Collect spillage.

Fire: In case of fire, use dry powder, foam, or carbon dioxide for extinction.

Storage:

Store in a cool, dry, well-ventilated place away from light. Maintain storage temperature between 5-35 deg C. Protect from freezing and keep packaging intact. Keep container tightly closed after use and consume promptly once opened.

Disposal:

Empty containers must not be used to store food or other goods prior to decontamination to prevent harm to human health and the environment. Dispose of packaging in accordance with national and local environmental regulations. Any hazards and losses caused by improper disposal shall be borne by the disposer.

Physical and chemical hazards: May cause contamination to water bodies and soil.

Health hazards: None

Environmental hazards: Prevent entry into drains to avoid polymer deposition and blockage of waterways.

3.Composition/Information on Ingredients

3.1 Product type:	Mixture	
3.2 Ingredients:		
Chemical name	**CAS NO.**	**Concentration % (w/w)**
Hydroxy acrylic polymer	Trade secret	46±5
Water	7732-18-5	45.5±5
Propylene glycol butyl ether	5131-66-8	7±3

3.Composition/Information on Ingredients		
2-Dimethylaminoethanol	108-01-0	1.5±0.5

4.First aid measures	
4.1 General measures:	Remove all contaminated clothing immediately.
4.2 Ingestion:	Do NOT induce vomiting. Seek medical attention immediately.
4.3 Eye contact:	Hold eyelids open and flush with plenty of warm water for at least 10 minutes. Consult an ophthalmologist.
4.4 Skin contact:	Wash immediately with plenty of soap and water. If skin reaction occurs, seek medical advice.
4.5 Inhalation:	If respiratory irritation occurs, seek medical advice.

5.Fire fighting measures	
5.1 Suitable extinguishing agents:	Carbon dioxide (CO2), foam, extinguishing powder. For large fires, use water spray.
5.2 Unsuitable extinguishing agents:	High-volume water jet.
5.3 Special hazards of substances or mixtures:	Combustion may release carbon monoxide, carbon dioxide, nitrogen oxides and trace amounts of hydrogen cyanide. Do not breathe fumes/smoke in case of fire or explosion.
5.4 Firefighting precautions protective equipment:	Firefighters must wear self-contained breathing apparatus. Do not allow contaminated fire-fighting water to enter soil, groundwater, or surface water.

6.Measures for accidental leakage	
6.1 Firefighting precautions protective equipment:	Wear protective equipment. Ensure adequate ventilation. Keep unauthorized persons away.
6.2 Environmental precautions:	Do not discharge into drains, wastewater, or soil.
6.3 Methods and material for containment and cleaning up:	Absorb with chemical absorbent material or dry sand if necessary, and store in tightly closed containers.

7.Operation and storage	
7.1 Precautions for safe handling:	Observe common chemical precautions during handling. Avoid contact with skin and eyes. Keep away from food, drink and tobacco. Wash hands before breaks and at the end of workday. Store work clothing separately. Change contaminated or soaked clothing.
7.2 Conditions for safe storage:	Keep containers tightly closed in a dry, well-ventilated place. To maintain product quality, strictly follow the storage conditions in our technical data sheet.

8.Exposure control / personal protection	
8.1 Exposure controls:	
Respiratory protection:	Respiratory protection equipment is required during spraying.
Hand protection:	Protective gloves are recommended.
Suitable material for protective gloves:	EN 374-3
Fluororubber – FKM:	Thickness ≥ 0.4 mm; Breakthrough time ≥ 480 min.
Butyl rubber – IIR:	Thickness ≥ 0.5 mm; Breakthrough time ≥ 480 min.
Nitrile rubber – NBR:	Thickness ≥ 0.35 mm; Breakthrough time ≥ 480 min.
Recommendation:	Contaminated gloves should be discarded.
Eye protection:	Wear protective goggles or face shield.
Body protection:	Wear appropriate protective clothing.
8.2 Engineering controls:	No exposure limit information is available under EC Directive 2006/121/EC.

9.Physical and chemical properties	
9.1 Appearance:	Milky white liquid
9.2 Odour threshold:	No data available.
9.3 pH value:	7-9
9.4 Ionicity:	Non-ionic
9.5 Initial boiling point:	Approx. 100℃

9.Physical and chemical properties	
9.6 Flash point:	95℃
9.7 Evaporation rate:	No data available.
9.8 Flammability:	Not applicable
9.9 Heat of combustion:	Not applicable
9.10 Relative vapour density:	No data available.
9.11 Density:	Approx. 1.06 g/cm³
9.12 Solubility in water:	Miscible
9.13 Surface tension:	No data available.
9.14 log Pow (n-octanol/water partition coefficient): partition coefficient): partition coefficient):	No data available.
9.15 Auto-ignition temperature: temperature: temperature:	Not applicable
9.16 Ignition temperature:	No data available.
9.17 Dynamic viscosity:	50-4000 mPa.s
9.18 Solid content:	46±2%
9.19 Other information:	The above data are not product specifications. Please refer to the Technical Data Sheet for specifications.

10.Stability and reactivity	
10.1 Chemical stability:	No decomposition if used according to specifications.
10.2 Hazardous decomposition products:	Flammable and harmful gases generated during thermal decomposition.
10.3 Possible hazardous reactions:	No hazardous reactions when stored and handled correctly.

11.Toxicity information
No toxicological studies are available on the product.
The following is toxicological data for these ingredients.
Hydroxy polyacrylate dispersion:

11.Toxicity information

Toxicity: No data available;

Irritation: No data available.

Propylene glycol butyl ether:

Toxicity: Dermal (Rat) LD50: > 2000 mg/kg

Inhalation (Rat) LD50: > 1997.718 mg/l/8hE

Oral (Rat) LD50: > 2000 mg/kg

Irritation: No data available.

2-(2-Butoxyethoxy)ethanol:

Toxicity: Dermal (Rat) LD50: > 2000 mg/kg

Oral (Rat) LD50: = 4500 mg/kg

Irritation: No data available.

12.Ecological information

12.1 Ecotoxicity:	No ecotoxicological studies are available for this product. Do not allow to enter drains, water courses or soil.
12.2 Persistence and degradability:	The product is not readily biodegradable.
12.3 Other adverse effects:	According to ecotoxicological data, the product is classified as non-hazardous to fish and daphnia.

13.Handling precautions

Dispose of in accordance with applicable national standards and national or local regulations.

Within the European Union, dispose of in accordance with the applicable provisions of the European Waste Catalogue (EWC).

Disposal methods:	Empty containers as completely as possible (e.g., by pouring, scraping, or draining until “drip-dry”). They may be sent to an appropriate collection point under existing chemical industry recycling schemes. Recycle containers in accordance with national regulations and

13.Handling precautions	
	applicable environmental requirements. Do not discharge waste via wastewater.

14.Transportation information	
14.1 Road and rail transport:	Not classified as dangerous goods for land transport, in accordance with ADG regulations.
14.2 Maritime transport:	Not classified as dangerous goods for sea transport, in accordance with IMDG regulations.
14.3 Air transport:	Not classified as dangerous goods for air transport, in accordance with IATA regulations.
14.4 Special precautions for user:	Not regulated as dangerous goods. Temperature must not exceed +40℃ and must not fall below +5℃. Keep away from food, acids and bases.

15.Regulatory information
Relevant safety, health and environmental regulations for the substance or mixture
Other provisions:
Complies with the following regulations:
Regulations on the Safety Management of Hazardous Chemicals, State Council Decree No. 591
GB/T 16483 Safety data sheet for chemical products - Content and order of sections
GB 13690 General rules for classification and hazard communication of chemicals
GB 30000.2-29 Rules for classification and labelling of chemicals
GB 15258 General rules for preparation of precautionary label for chemicals

16.Other information
To the best of our knowledge, the information provided in this Safety Data Sheet is accurate as of the date of publication. The information given is designed only as guidance for safe handling, use, processing, storage, transportation, disposal, and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.